

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 2: CELL: THE BUILDING BLOCK OF LIFE [SET B - ADVANCED HOTS]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH02/SET-B****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET B - HIGHER ORDER THINKING SKILLS):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Advanced Conceptual and Numerical MCQs with single correct option.
3. **Section B (Q16 – Q20):** Critical Assertion-Reasoning based questions testing analytical clarity.
4. **Section C (Q21 – Q25):** Case-Study / Practical Experimental Data-interpretation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: ADVANCED & NUMERICAL MCQS (Q.1 TO Q.15)**[15 MARKS]****Q.1 Why are most biological cells microscopic in size rather than growing into giant macroscopic entities?**

- (A) They lack sufficient DNA to divide (B) As a cell grows, its volume increases faster than its surface area, limiting nutrient diffusion across the membrane
- (C) Large cells are crushed by atmospheric air pressure (D) Cell organelles cannot replicate inside large volumes

Q.2 A student places peeled raw potato cups in three troughs: Trough A (water, empty cup), Trough B (water, cup filled with 10% sugar solution), and Trough C (boiled potato cup, filled with 10% sugar solution). Water collects inside the cup in:

- (A) Trough A only (B) Trough B only
- (C) Trough B and Trough C (D) Trough C only

Q.3 Which sequence correctly traces the intracellular pathway of a newly synthesized digestive enzyme from its site of manufacture to secretion outside the cell?

- (A) Golgi apparatus → Ribosome → RER → Plasma membrane (B) Rough ER → Transport vesicle → Golgi apparatus → Secretory vesicle → Plasma membrane
- (C) Smooth ER → Lysosome → Mitochondria → Plasma membrane (D) Nucleus → Cytoplasm → Vacuole → Plasma membrane

Q.4 Why does a plant cell not undergo lysis (bursting) when placed in pure distilled water, unlike a human red blood cell?

- (A) Plant cells do not absorb water in hypotonic solutions (B) The rigid cellulose cell wall exerts inward wall pressure equal and opposite to turgor pressure
- (C) Plant cytoplasm contains zero dissolved solutes (D) Plant cell membranes lack aquaporin channels

Q.5 Which cellular organelle is intimately involved in membrane biogenesis (the synthesis of lipids and proteins required to build new cell membranes)?

- (A) Lysosomes and Peroxisomes (B) Endoplasmic Reticulum (RER for proteins, SER for lipids)
- (C) Centrioles and Centrosomes (D) Plastids and Vacuoles

Q.6 The inner mitochondrial membrane contains a high proportion of cardiolipin and is deeply folded into cristae. In which cells would you expect to find the highest density of cristae?

- (A) Dormant plant seed cells
- (B) Sperm tail cells and avian flight muscle cells
- (C) Subcutaneous adipose fat cells
- (D) Dead cork bark cells

Q.7 The primary structural component of the fungal cell wall that differs chemically from plant cellulose is:

- (A) Peptidoglycan
- (B) Chitin
- (C) Pectin
- (D) Glycogen

Q.8 Amoeba engulfs food particles from its external environment by the inward folding of its flexible plasma membrane. This process is termed:

- (A) Plasmolysis
- (B) Endocytosis (Phagocytosis)
- (C) Exosmosis
- (D) Facilitated diffusion

Q.9 Which of the following cellular structures lacks any surrounding membrane and is responsible for assembling ribosomal subunits inside the nucleus?

- (A) Nuclear envelope
- (B) Nucleolus
- (C) Golgi cisterna
- (D) Peroxisome

Q.10 What happens when green tomatoes ripen and transform into bright red tomatoes?

- (A) Chloroplasts transform into chromoplasts as chlorophyll degrades and carotenoids/lycopene accumulate
- (B) Leucoplasts transform into chloroplasts
- (C) Chromoplasts convert into vacuoles
- (D) Cell walls dissolve completely

Q.11 A cell with diploid chromosome number $2n = 16$ undergoes mitotic division. What will be the chromosome number in each of the two resulting daughter cells?

- (A) 8
- (B) 16
- (C) 32
- (D) 4

Q.12 If a germ cell with $2n = 16$ chromosomes undergoes meiosis to form gametes, how many daughter cells are produced and what is their chromosome count?

- (A) 2 cells with 16 chromosomes each
- (B) 4 cells with 8 chromosomes each
- (C) 4 cells with 16 chromosomes each
- (D) 2 cells with 8 chromosomes each

Q.13 Which organelle contains hydrolytic digestive enzymes that function optimally in an acidic luminal pH (approx 4.5 - 5.0)?

- (A) Peroxisome
- (B) Lysosome
- (C) Mitochondrion
- (D) Ribosome

Q.14 The single membrane bounding the large central sap vacuole in plant cells is specifically designated as the:

- (A) Tonoplast
- (B) Plasmalemma
- (C) Pellicle
- (D) Crista

Q.15 Chromatin material is biochemically constituted of:

- (A) DNA and histone proteins
- (B) RNA and lipids only
- (C) Carbohydrates and minerals
- (D) Pure double-stranded RNA

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 (C) Assertion (A) is true but Reason (R) is false.
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): Mitochondria and chloroplasts are described as semi-autonomous organelles.

Reason (R): They contain their own circular DNA genomes and 70S ribosomes, capable of self-replication and protein synthesis.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): The plasma membrane is completely rigid and impermeable to water molecules.

Reason (R): The plasma membrane is composed of a continuous bilayer of cellulose fibers.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) Both (A) and (R) are false.

Q.18 Assertion (A): Lysosomes are responsible for autolysis of worn-out cells.

Reason (R): When a cell is severely damaged or aging, lysosomes rupture and release powerful digestive enzymes that digest the cellular components.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Meiosis is essential for maintaining constant chromosome number across successive generations in sexually reproducing species.

Reason (R): Meiosis halves the chromosome number in gametes, which is subsequently restored upon fertilization.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): Virchow's addition to the cell theory contradicted Schleiden and Schwann's original propositions.

Reason (R): Schleiden and Schwann claimed that all living things are composed of cells, whereas Virchow discovered the nucleus.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) Both (A) and (R) are false.

SECTION C: EXPERIMENTAL DATA & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Microscopic Investigation of Plasmolysis in *Rhoeo* Leaf Epidermis

Students peel the lower purple epidermis of a fresh *Rhoeo discolor* leaf, which contains anthocyanin dye in its cell sap vacuoles. They mount the peel in a drop of water on Slide 1 and examine it under 400x magnification, noting bright purple protoplasts fully filling the polygonal cell walls. Next, they add 3 drops of concentrated 20% sucrose solution to Slide 2. Within 5 minutes, the purple protoplast contracts into an oval blob in the center, leaving clear gaps between the protoplast and the cell wall. Finally, they flush Slide 2 with excess fresh distilled water.

Q.21 Why is *Rhoeo discolor* epidermis chosen specifically for this osmosis demonstration?

- (A) It lacks a cell wall
(B) Its vacuoles contain natural purple anthocyanin pigment, making the protoplast boundary distinctly visible without artificial staining
(C) It cannot undergo osmosis
(D) Its cells are dead

Q.22 The withdrawal of the purple protoplast away from the cell wall observed on Slide 2 is called:

- (A) Endosmosis
(B) Plasmolysis
(C) De-plasmolysis
(D) Cytokinesis

Q.23 What fluid occupies the space between the shrunken protoplast and the cell wall on Slide 2?

- (A) Pure vacuum
(B) Air bubbles
(C) The external hypertonic sucrose solution
(D) Pure water generated by the cell

Q.24 When excess distilled water is added to flush the sucrose away from Slide 2, what phenomenon is observed?

- (A) The cells burst immediately
(B) De-plasmolysis occurs as water re-enters by endosmosis, restoring protoplast turgidity against the wall
(C) The cells die permanently
(D) The purple pigment disappears forever

Q.25 If the *Rhoeo* leaf peel had been boiled in hot water for 5 minutes prior to adding sucrose solution, what would be seen under the microscope?

- (A) Plasmolysis would occur twice as fast
(B) No plasmolysis would occur because boiling destroys living semi-permeable membranes
(C) Cells would double in volume
(D) Vacuoles would turn bright green

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & IN-DEPTH EXPLANATIONS (SET B)

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.1	(B)	Surface area increases as r^2 while volume increases as r^3 . A small cell maintains a high surface area to volume ratio (SA/V) necessary for rapid metabolic transport.
Q.2	(B)	Water enters cup B by endosmosis across living selectively permeable potato cell membranes. In cup C, boiling killed cells, destroying osmotic selective permeability.
Q.3	(B)	Proteins synthesized on ribosomes of the Rough ER are shuttled via transport vesicles to the Golgi for modification/sorting, then packaged into secretory vesicles for exocytosis.
Q.4	(B)	As water enters a plant cell, turgor pressure builds up; the rigid cellulose wall exerts equal inward wall pressure, halting further osmotic influx before bursting.
Q.5	(B)	RER synthesizes membrane proteins and SER synthesizes membrane lipids (phospholipids and cholesterol), together driving membrane biogenesis.
Q.6	(B)	Cells with extreme, continuous energetic demands (such as avian flight muscles and motile sperm) require maximum cristae surface area to pack respiratory electron transport chains and ATP synthases.
Q.7	(B)	Fungi possess cell walls constructed from chitin (a polymer of N-acetylglucosamine), whereas plants use cellulose.
Q.8	(B)	The fluidity and flexibility of the lipid bilayer allows amoeba to engulf external particulate food through endocytosis/phagocytosis.
Q.9	(B)	The nucleolus is a dense, non-membrane-bound subnuclear region where ribosomal RNA (rRNA) is transcribed and combined with proteins into ribosomal subunits.
Q.10	(A)	During ripening, chloroplasts structurally reconfigure into chromoplasts: green chlorophyll is broken down while red carotenoid pigments (like lycopene) are synthesized.
Q.11	(B)	Mitosis is equational division. The parent chromosome number is exactly maintained ($2n = 16$ in both daughter cells).
Q.12	(B)	Meiosis undergoes two successive nuclear divisions producing 4 haploid gametes, each having half ($n = 8$) the chromosome count of the parent cell.
Q.13	(B)	Lysosomes maintain an acidic interior using proton pumps, optimizing the activity of acid hydrolases (proteases, nucleases, lipases).
Q.14	(A)	The tonoplast is the semi-permeable membrane enclosing the plant central vacuole, maintaining hydrostatic turgor pressure.
Q.15	(A)	Chromatin consists of DNA wound around octamers of basic histone proteins to form nucleosome fibers.
Q.16	(A)	Because they carry their own genetic machinery and divide by binary fission independently of nuclear division, they are semi-autonomous.
Q.17	(D)	Both statements are completely false. The plasma membrane is a flexible, fluid mosaic of phospholipids and proteins, highly permeable to water via aquaporins.
Q.18	(A)	Lysosomal rupture during programmed cell death or necrotic injury releases hydrolytic enzymes that break down all macromolecular components of the dead cell.
Q.19	(A)	If gametes were diploid ($2n$), fertilization would double chromosomes every generation ($4n, 8n$). Meiosis produces haploid (n) gametes so $n + n = 2n$ is conserved.
Q.20	(D)	Both are false. Virchow expanded (did not contradict) cell theory by adding 'Omnis cellula-e cellula' (cells arise from pre-existing cells). Robert Brown discovered the nucleus.
Q.21	(B)	Anthocyanin pigments dissolve in vacuolar sap, giving a vibrant violet hue that makes shrinking of the living protoplast easily visible.

Q. No.	Correct	In-Depth Conceptual & Numerical Solution
Q.22	(B)	Plasmolysis occurs when exosmosis draws water out of the central vacuole in a hypertonic medium, pulling the plasma membrane away from the wall.
Q.23	(C)	The cellulose cell wall is fully permeable to water and small solutes (like sucrose), allowing the external sucrose solution to occupy the gap.
Q.24	(B)	De-plasmolysis is the reversal of plasmolysis when cells are returned to a hypotonic medium before permanent damage occurs.
Q.25	(B)	Boiling denatures proteins and disintegrates the lipid bilayer, killing cells and eliminating selective permeability.